

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	

Application of Concepts	25	Effectively applies relevant concepts to	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
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ASSESSMENT TOOL 2 – Coding Challenge

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO5-AT2 Coding Challenge	Assessment:	ASSESSMENT TOOL 2 – Coding Challenge
Weightage:	40 % of CIA	Total Marks:	100
CO4 Statement:	To understand the concept of supervised and unsupervised methods in machine learning		
Student Name:	M V RAGHAVAN	Reg. No.:	192424373
Batch	2024	Date:	27-08-2026

Code of Conduct

I M V RAGHAVAN (192424373) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M V RAGHAVAN

		analyze the case deeply				
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Code:

```
import numpy as np
from sklearn.linear_model import LinearRegression, Ridge
np.random.seed(10) n = 100
X1 = np.random.randn(n)
X2 = X1 + np.random.randn(n) * 0.1
X3 = np.random.randn(n)
y = 3*X1 + 2*X3 + np.random.randn(n)
X = np.column_stack([X1, X2, X3])
names = ["X1", "X2", "X3"]
print("VIF values:")
for i in range(X.shape[1]):
    other = np.delete(X, i, axis=1)
    r2 = LinearRegression().fit(other, X[:, i]).score(other, X[:, i])
    vif = 1 / (1 - r2)
    print(names[i], round(vif, 2))
def bootstrap(model, X, y, B=200):
    coefs = []
    for _ in range(B):
        idx = np.random.choice(len(y), len(y), replace=True)
        model.fit(X[idx], y[idx])
        coefs.append(model.coef_)
    return np.std(coefs, axis=0)
print("\nOriginal coefficient SD:")
print(bootstrap(LinearRegression(), X, y))
X_reduced = X[:, [0, 2]]
print("\nAfter removing X2:")
print(bootstrap(LinearRegression(), X_reduced, y))
print("\nAfter Ridge regularization:")
print(bootstrap(Ridge(alpha=1), X, y))
```

Output:

... VIF values:

X1 99.99

X2 99.92

X3 1.01

Original coefficient SD:

[1.01649163 1.01302447 0.09869164]

After removing X2:

[0.08350453 0.09163825]

After Ridge regularization:

[0.3135496 0.31891239 0.10192778]

